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RESEARCH STATEMENT

My research is positioned at the intersection of electrical engineering, embedded computer science, robotics, artificial intelligence, and intelligent energy systems. As a Doctor in Electrical Engineering and a Postdoctoral Fellow within the Robotics and Internet of Things Unit at Prince Sultan University, my work focuses on designing reliable, explainable, and deployable intelligent systems for real-world engineering applications. My research trajectory has evolved from hybrid electric vehicle energy management and battery modeling toward autonomous UAV navigation, GNSS-denied localization, embedded AI, robotics, and IoT-based sensing systems.

A defining characteristic of my research profile is the integration of theoretical modeling, data-driven learning, sensor fusion, simulation, and experimental validation. I aim to develop methods that are mathematically grounded, computationally efficient, and suitable for implementation on embedded or robotic platforms. My work therefore combines algorithm design with practical engineering constraints such as sensor uncertainty, computation time, hardware limitations, environmental variability, and real-time operation.

RESEARCH VISION

My long-term research vision is to contribute to the development of autonomous and intelligent systems that can operate safely and accurately in uncertain, constrained, and infrastructure-limited environments. I am particularly interested in UAVs, mobile robots, embedded AI systems, and energy-aware intelligent platforms. These systems require robust perception, localization, prediction, control, and decision-making methods that remain reliable when conventional assumptions fail, such as when GNSS signals are degraded or unavailable, sensor measurements are noisy, or environmental maps are incomplete.

To address these challenges, I pursue a research strategy that combines physics-based modeling, machine learning, deep learning, and sensor fusion. I consider hybrid approaches essential because purely model-based methods may struggle with complex nonlinear environments, while purely data-driven models may lack interpretability, robustness, or generalization. My objective is to build methods that benefit from both domains: the structure and reliability of engineering models and the adaptability of modern artificial intelligence.

RESEARCH AREAS AND CONTRIBUTIONS

Research direction	Main contributions and expertise
Autonomous UAV navigation	GNSS-denied localization, terrain-aided navigation, sensor fusion, DEM-based positioning, ROS 2, PX4, Gazebo, and real/simulated UAV datasets.

Research direction	Main contributions and expertise
Embedded AI and robotics	CNN-LSTM-Attention, KAN-based architectures, GRU predictors, embedded decision-making, drone-to-drone tracking, and intelligent robotic systems.
Energy and battery systems	Hybrid electric vehicle energy management, Li-ion battery and supercapacitor modeling, SOC/SOH estimation, optimization, and HIL validation.
IoT and intelligent sensing	Sensor-based monitoring, public-lighting fault management, smart mailbox systems, biomedical sensing, and embedded hardware integration.
Open-source research infrastructure	Dataset generation, reproducible simulation environments, georeferenced terrain preprocessing, and benchmark frameworks for autonomous navigation.

UAV LOCALIZATION AND NAVIGATION IN GNSS-DENIED ENVIRONMENTS

My recent research has focused strongly on autonomous UAV localization and navigation under degraded or unavailable GNSS conditions. This problem is critical for real-world UAV operations in urban, mountainous, indoor, contested, or infrastructure-limited environments. In these scenarios, conventional satellite-based navigation may be unreliable, and the UAV must estimate its position using onboard sensors, environmental maps, and intelligent fusion algorithms.

I have contributed to the development of UAV localization frameworks that integrate inertial measurements, LiDAR or altitude-related information, barometric data, visual or non-visual sensing, and georeferenced terrain information such as DEM, DSM, and GeoTIFF maps. My work includes terrain-aided navigation, TERCOM-inspired matching, EKF-based fusion, residual learning, and AI-enhanced localization models. These approaches aim to reduce localization drift and improve position recovery during GNSS-denied intervals.

A major direction of this work is the design of hybrid learning and filtering architectures. I have worked on approaches involving CNN-LSTM-Attention models, Kolmogorov-Arnold Network-based components, GRU predictors, XGBoost residual learning, and EKF-based correction mechanisms. These models are designed to exploit temporal sensor patterns, elevation information, motion dynamics, and terrain constraints. The goal is to provide localization methods that are accurate, robust, and suitable for autonomous navigation in realistic outdoor environments.

OPEN-SOURCE DATASET GENERATION AND SIMULATION-BASED VALIDATION

A second major research direction concerns the creation of reproducible research infrastructure for UAV localization and tracking. I have contributed to open-source frameworks that use ROS 2, PX4, Gazebo, QGIS, Blender, and georeferenced terrain data to generate UAV datasets and validate autonomous navigation algorithms. These frameworks are important because progress in GNSS-denied navigation depends not only on algorithm design, but also on high-quality datasets, realistic simulation environments, and transparent evaluation protocols.

My work in this direction includes UAV mission simulation, georeferenced data collection, synchronized sensor recording, trajectory generation, DEM/DSM-based terrain preprocessing, and experimental validation workflows. By combining simulation and real-world-inspired sensing modalities, I seek to reduce the gap

between algorithmic development and practical deployment. I am particularly interested in building benchmark environments that allow researchers to compare methods under controlled GNSS-denial scenarios, different terrain profiles, sensor noise levels, and flight trajectories.

ROBOTICS, TRACKING, AND EMBEDDED ARTIFICIAL INTELLIGENCE

Beyond UAV localization, I am interested in intelligent robotic systems that combine sensing, prediction, and decision-making. My work includes aerial target tracking, drone-to-drone tracking, trajectory prediction, and the integration of Kalman filtering with neural sequence models. These systems require fast inference, temporal reasoning, and robustness to motion uncertainty, occlusion, and dynamic environmental conditions.

Within RIOTU, I have also participated in research related to quadruped robotics, embedded AI, and autonomous decision-making. My broader objective is to contribute to robotic systems that can perceive their environment, estimate their state, predict future motion, and act reliably in complex conditions. This requires collaboration across robotics, control, computer vision, embedded systems, machine learning, and real-time software engineering.

EMBEDDED SYSTEMS, WIRELESS COMMUNICATION, AND INDUSTRIAL AI APPLICATIONS

In addition to my academic research, I have developed strong practical expertise in the design and programming of embedded systems for robotics, Internet of Things applications, and artificial-intelligence-based industrial solutions. This experience strengthens my ability to connect algorithmic research with real embedded implementation, sensor integration, hardware prototyping, and field-oriented validation.

During my industrial R&D experience, particularly at Novel-Ti, I worked on AI-enabled applications involving facial recognition, motion detection for children's interactive games, and computer-vision-based detection of football balls for VAR-related applications. These projects allowed me to combine embedded programming, image processing, real-time constraints, and applied artificial intelligence in practical product-oriented contexts.

I have also contributed to research and industrial projects involving wireless communication systems for data transfer between distributed devices. My practical skills include PCB design using Altium Designer and the use of wireless technologies such as Zigbee, LoRa, ESP32, and SIM8008. These technologies were applied in monitoring-oriented systems, including temperature and humidity measurement in beehive environments and a thermistor-based food monitoring project that allowed restaurant chefs to follow the temperature evolution of food during preparation and storage.

This experience also gives me a practical understanding of the operational range, advantages, and limitations of wireless communication modules, including short-range Zigbee communication, long-range LoRa transmission, cellular communication modules, and ESP32-based connected systems. Such expertise is directly relevant to smart monitoring and cloud-connected data acquisition applications, including agricultural and food-production chains where reliable sensing, reduced human mobility, remote monitoring, and accurate data storage are essential. For example, these concepts can support systems that monitor olive oil quality and environmental parameters throughout the production chain.

ENERGY MANAGEMENT, BATTERY MODELING, AND INTELLIGENT POWER SYSTEMS

My doctoral research established my foundation in hybrid electric vehicle energy management, with a focus on systems combining Li-ion batteries and supercapacitors. This work involved battery and supercapacitor modeling, state-of-charge estimation, recursive least squares identification, optimization methods, power management strategies, and hardware-in-the-loop validation using MATLAB, Simulink, dSPACE, and

embedded test benches. This experience shaped my ability to connect theory, simulation, and experimental validation.

This research line continues through my work on battery state estimation and health monitoring. I am interested in AI-enhanced battery modeling, SOC and SOH prediction, uncertainty-aware estimation, and hybrid neural architectures such as KAN-LSTM models. These methods are important for electric vehicles, renewable energy storage, smart grids, and energy-aware robotic systems. My long-term objective in this domain is to develop reliable and interpretable models that support safer and more efficient energy management.

METHODOLOGICAL APPROACH

My research methodology is based on a complete engineering cycle. I begin by defining the problem and identifying the physical constraints, sensor limitations, and system requirements. I then develop mathematical models, design algorithms, implement them in software, and validate them through simulation or experimental datasets. I place strong emphasis on quantitative evaluation using metrics such as RMSE, MAE, R-squared, inference time, robustness under noise, trajectory deviation, and computational complexity.

I also value reproducibility and transparency. For this reason, I support the use of open-source software, documented pipelines, benchmark datasets, clear ablation studies, and well-defined evaluation protocols. In my view, strong research must provide not only good results, but also a clear explanation of why the proposed method works, under which conditions it performs well, and where its limitations remain.

FUTURE RESEARCH AGENDA

My future research agenda will continue to develop around three complementary directions. The first is robust autonomous localization for UAVs and mobile robots under GNSS-denied conditions. I aim to develop hybrid frameworks that combine terrain-aided navigation, sensor fusion, physics-guided learning, uncertainty estimation, and real-time embedded deployment. A key objective is to move from offline evaluation toward real-time validation on robotic platforms.

The second direction is embedded AI for autonomous decision-making. I plan to investigate lightweight and interpretable learning architectures for robotics, including KAN-based models, attention mechanisms, reinforcement learning, and agentic decision-making frameworks. My interest is not only to improve accuracy, but also to improve reliability, explainability, and deployment feasibility on embedded hardware such as Jetson platforms, microcontrollers, and edge devices.

The third direction is intelligent energy-aware autonomy. Future autonomous systems must not only localize and navigate effectively, but also manage their energy resources efficiently. I therefore aim to connect my expertise in electric vehicle energy management and battery health monitoring with robotics and UAV applications. This includes battery-aware mission planning, energy-constrained navigation, health-aware power management, and intelligent monitoring systems for mobile robots and UAVs.

COLLABORATION, MENTORING, AND SCIENTIFIC IMPACT

My research activity is strongly connected to collaboration, mentoring, and scientific dissemination. I value interdisciplinary collaboration because the challenges addressed in autonomous systems, robotics, IoT, and energy management require expertise from multiple domains. I have contributed to research projects, journal publications, conference papers, open-source software, student mentoring, and scientific activities. These experiences have strengthened my ability to work in teams, supervise students, support research writing, and contribute to high-impact academic outputs.

As a researcher and university teacher, I also consider mentoring an essential part of my research mission. I aim to guide students and young researchers in defining research questions, building experimental pipelines, analyzing results, writing scientific papers, and developing professional research habits. My goal is to contribute not only through my own research outcomes, but also by helping develop the next generation of engineers and researchers in robotics, embedded AI, energy systems, and intelligent autonomous technologies.

CONCLUSION

In summary, my research seeks to bridge electrical engineering, computer science, artificial intelligence, robotics, embedded systems, wireless sensing, and energy systems. My work is motivated by the need to create intelligent technologies that are accurate, reliable, reproducible, and deployable in real-world environments. Through research, collaboration, supervision, and open-source development, I aim to contribute to autonomous systems, smart monitoring, energy-aware intelligence, and embedded AI solutions that address practical engineering challenges.